Dockerizing the QR Code Generator

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GitHub repository https://github.com/tkg1121/qr-code-generator

DockerHub image https://hub.docker.com/r/tkg1121/qr-code-generator-app

Successful GitHub Actions run https://github.com/tkg1121/qr-code-generator/actions/runs/39527047973

Reflection

Dockerizing the QR Code Generator helped me better understand how applications can be packaged and run in different environments. Since the Python application was already provided, most of my focus was on configuring Docker and making sure everything worked together. I understood the general purpose of containers before this assignment, but actually building and running one helped connect some of those concepts (it's one thing to understand what a container does, and another to figure out why yours isn't behaving the way you expected).

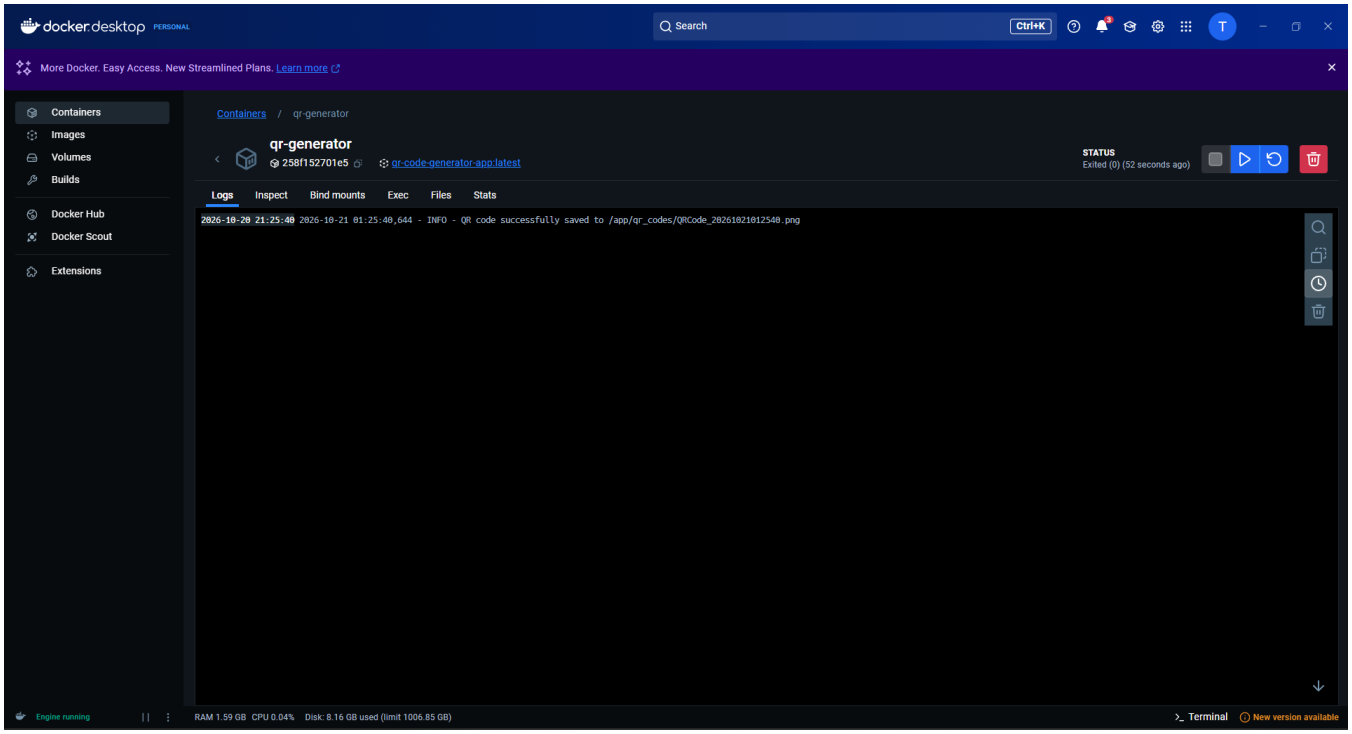
One thing I had to think through was how files move between the container and the host machine. The application could generate QR codes successfully, but that doesn't necessarily mean the files would be accessible outside the container. Using bind mounts helped me understand how to preserve those files, while running the application as a non-root user reinforced the importance of file permissions (because apparently getting the application to work isn't enough if it can't write to its own directory). I also liked being able to change settings through environment variables without rebuilding the entire image.

Another thing I learned was how to interpret the container's lifecycle. Initially, seeing a container exit might seem like something went wrong, but this application is only supposed to generate a QR code and finish. An Exited (0) status is actually the expected result. This reinforced that checking whether a container is running isn't always the best way to determine whether an application worked. Reviewing the logs and verifying the generated QR code provided much better evidence. GitHub Actions was also useful for automating the tests, building the image, and checking the output (which is much nicer than manually repeating everything and hoping I didn't miss a step).

Overall, I feel more comfortable reading and writing Dockerfiles, understanding how ENTRYPOINT and CMD work together, and publishing images to DockerHub. I still want more practice troubleshooting containers and working with Linux volume permissions. The assignment also helped me better understand the difference between having Docker installed and having the Docker engine actually running (two things that sound like they should go hand in hand, but apparently not always).

Container and Workflow Screenshots

Docker Desktop logs show the QR image was saved successfully. Exited (0) is the expected result for this one-run application.



GitHub Actions passed the automated tests, Docker build, and container output checks.

